LEI TIANYOU

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PROFILE SUMMARY

GDipSA (NUS-ISS) student focused on LLM agent engineering. Built a LangGraph publishing agent end to end — graph orchestration, terminology auditing to 81.8% consistency, and cost-aware model selection — and shipped it to a live deployment. Draws on machine learning benchmarking and Java/Spring Boot delivery.

SKILLS

- Machine Learning: Deep transfer learning (CNN); gradient boosting & ensemble methods (CatBoost, XGBoost); end-to-end ML workflow with SHAP interpretability
- Programming: Python, Java, MySQL (Proficient); JavaScript, HTML, CSS, Kotlin, R, C++ (Familiar)
- Full-Stack Web: Spring Boot, JPA/Hibernate (ORM), ReactJS, Django
- Business Analysis: ACCA Diploma in Accounting and Business, industry research
- Languages: English (IELTS 7.5, 2025), Chinese (Native), Japanese (Basic)

EDUCATION

National University of Singapore **Jul 2025 - Present**

Graduate Diploma in Systems Analysis (GDipSA)

- Coursework: Machine Learning, Agentic AI System, Web Application Development, Mobile Application Development, DevSecOps, Big Data

Chongqing University **Sep 2022 - Jun 2026**

Bachelor of Management, Administration Management

- Designed and executed the ML study within the degree thesis, predicting internal and external political efficacy from European Social Survey Round 11 data across ~39,000 samples in 23 European countries

PROJECTS

Sole Developer **Jul 2026 - Present**

Bilingual Publication-to-Markdown Publishing Agent | github.com/elenskye/pub2md-agent

- Built a LangGraph agent that converts PDFs into publication-ready bilingual Markdown notes, reaching 81.8% terminology consistency
- Cut roughly 3× cost and 6× latency by selecting Instruct over Thinking vision models for transcription, where step-by-step reasoning added nothing
- Shipped a live web service on a DigitalOcean droplet (nginx, gunicorn, TLS), scoping Django to a thin shell so agent iterations never touch the web layer

Android Developer **Jun 2026 - Jul 2026**

SimpleWell — AI-Assisted Wellness Tracking App

- Built Android login and registration in Kotlin — ViewModel state management, form validation and JWT session handling — merged via pull requests into a 7-person team repository
- Negotiated request and error-response contracts with the Spring Boot backend owner, and covered validation and error mapping with unit tests that kept auth verifiable before the backend was ready
- Contributed to the early design of the team's FastAPI AI service, co-designing an LLMProvider protocol layer so services depend on the interface rather than the DeepSeek SDK

Co-Author **Dec 2025 - May 2026**

Thesis — Key Predictors of Political Efficacy Based on ML

- Engineered a reproducible modelling pipeline in Python and R over 39,000 European Social Survey responses, applying complete-case cleaning and categorical encoding
- Benchmarked 11 models — tree-based ensembles, gradient boosting, SVR, KNN and a neural network — and selected CatBoost on R^2 (0.373 & 0.424), MAE and RMSE
- Ranked feature importance with SHAP mean absolute values and dependence plots

Team Leader **Dec 2025 - Jan 2026**

Fruit Image Classifier — CNN with Transfer Learning

- Led the Version 2 and final optimisation iterations for a 5-person team building a four-class fruit image CNN, applying MobileNetV2 transfer learning to reach 0.92 accuracy and 0.87 F1
- Validated robustness through a 5-configuration ablation and 3 random seeds, lifting validation macro-F1 from 0.901 to 0.932

Backend Developer **Sep 2025 - Nov 2025**

Full-Stack Shopping-Cart Web Application

- Implemented RESTful API services with two teammates on a Spring Boot backend, persisting data through JPA/Hibernate ORM over a SQL database
- Defined API contracts with a 2-person ReactJS frontend team and integrated both layers into one e-commerce application spanning 6 core features

EXPERIENCE

Business Centre Intern **Jan 2024 - Feb 2024**

CITIC Securities, Hubei Branch

- Produced industry research briefs by synthesising internal sell-side research coverage
- Led a 3-member team to design diversified investment strategies tailored to client needs
- Automated portfolio-return analysis with 2 scripts in Python and Excel VBA, and designed the reporting workflow feeding the final investment decision

Marketing Intern **Jul 2023 - Aug 2023**

Gaodun Education, Chongqing Headquarters

- Co-ran a Chongqing University seminar series to full enrolment, channelling attendee feedback into later sessions
- Recruited 10+ teams for the EY ESG Challenge through campus outreach and briefing sessions, several of which placed
- Co-managed a 350-member freshman community across campus promotion channels, sustaining active peer Q&A and resource sharing